# Computer Science & IT

MATHS 5-6 Question

37 mark cut off

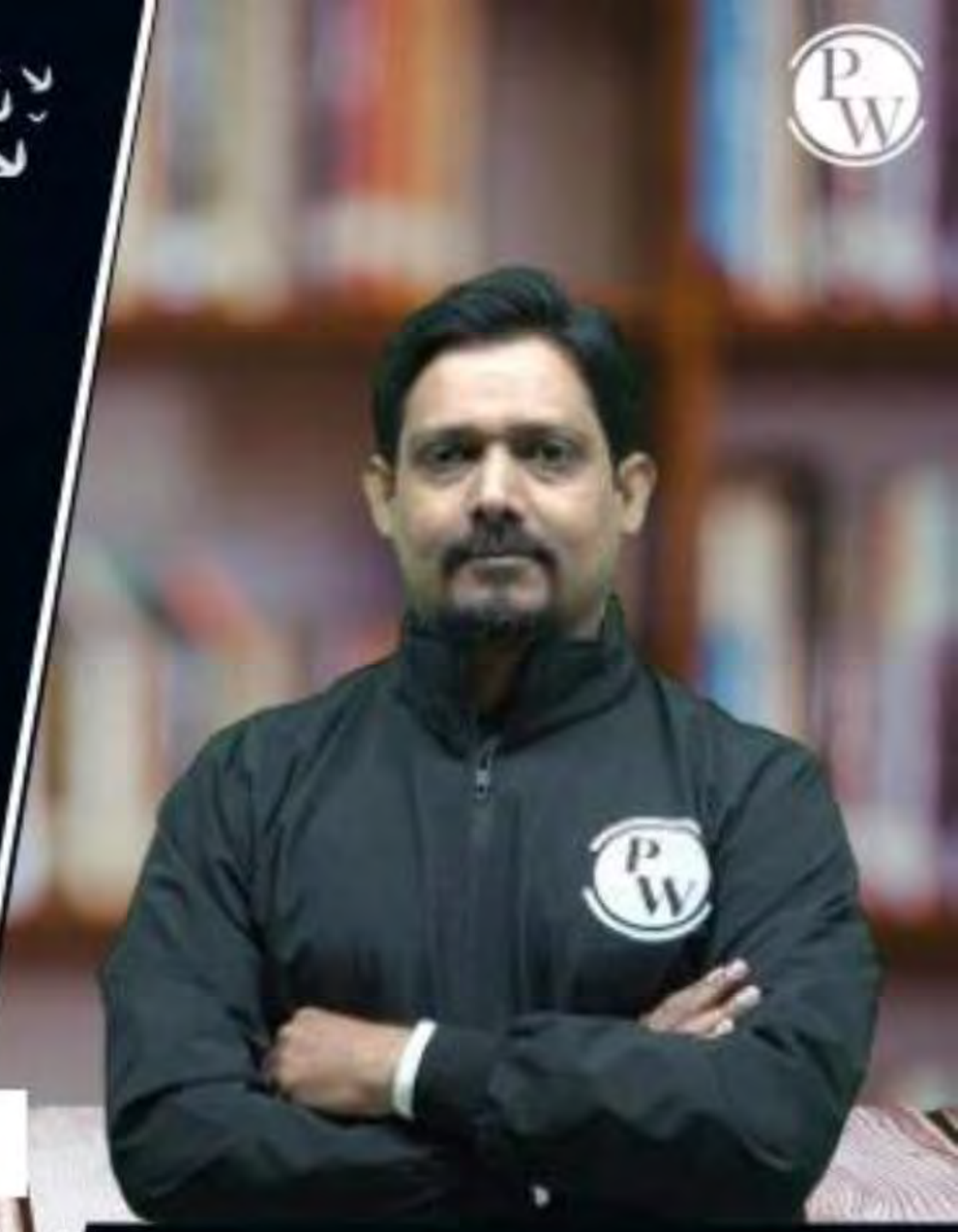
43 marks



# Data Structure

## Tree

Lecture No. 02



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# Recap of Previous Lecture

Maths + C-programming
+ Digital logic



Topic — Tree Terminology

Topic — forest . n total vertex p ⇒ No. of edges n-p

Topic — degree

1 Graph : No edges connected to a Node

2 degree of tree No. of children a Node is having

default

Topic

Topic

Slide

# Topics to be Covered



- Topic — k-ary tree
- Topic — Binary tree
- Topic — Height of Binary tree / No of Nodes
- Topic — k-ary tree No of Nodes (Minimum Height / Max height)
- Topic — Counting of unlabeled Node

Slide



# Topic : Tree

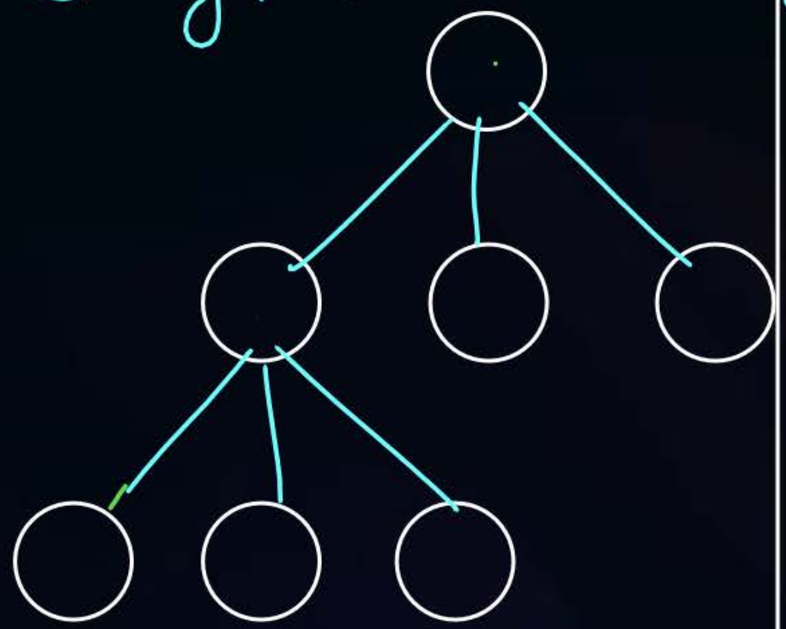
K ary tree
Binary tree
Height of binary tree
Binary tree theorem
Special tree
Counting unlabeled node



# Topic : Tree

K-ary tree is a tree in which each node is having 0 or k children

3-ary tree

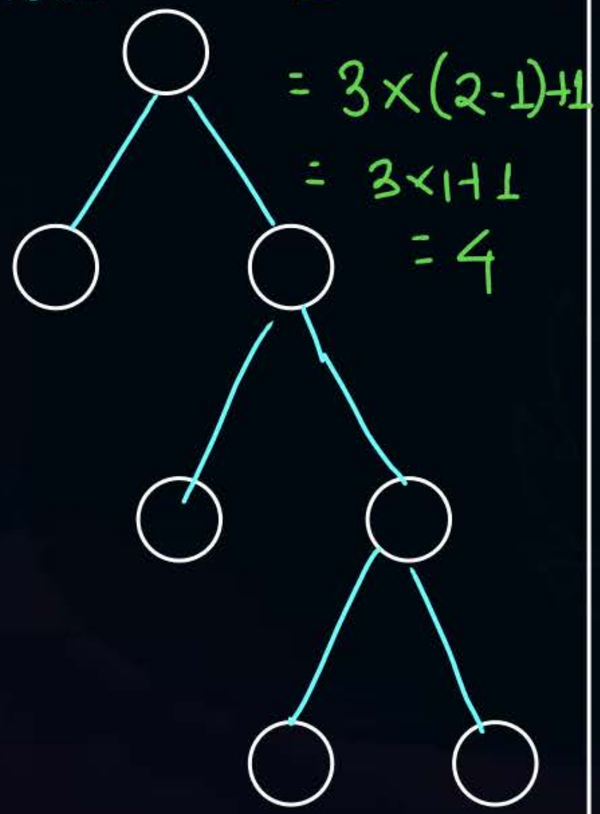


$L = I(k-1) + 1$

$L = 2 \times (3-1) + 1 =$

$2 \times 2 + 1 = 5$

2-ary tree



$L =$

$= 3 \times (2-1) + 1$

$= 3 \times 1 + 1$

$= 4$

Special

General tree

Slide



# Topic : Tree

**Theorem :-** if L is No. of Leaves and I is No. of No. of Internal Node in k-ary tree

$$L = I(k-1)+1$$

Slide



# Topic : Tree

degree of a Node is No edges it connected, L = # of leaves, I = # of internal Node

K-ary degree of leaf Node : 1

degree of internal Node : K+1

Except root Node

No of Nodes : L+I

No of edges : L+I-1

degree of a Node is No. of children a Node is having. (L is No of Leaves, I is No of internal Node)

degree of leaf Node : 0

degree of internal Node : K

No. of Nodes = L+I

No. of edges = L+I-1

Slide



# Topic : Tree

Handshaking

$$\sum_{i=1}^{n} deg(v_i) = 2|E|$$

Adjustment of Root

$$L \times 1 + I(k+1) - 1 = 2(L+I-1)$$

$$L + Ik + I - 1 = 2L + 2I - 2$$

$$\Rightarrow 2L - L = Ik + I - 1 - 2I + 2$$

$$\therefore L = Ik - I + 1$$

$$L = I(k-1) + 1$$

Slide

Sum of degree = total No. of edge

$$L \times 0 + Ik = (L + I - 1)$$

$$L = Ik - I + 1$$

$$\boxed{L = I(k-1) + 1}$$

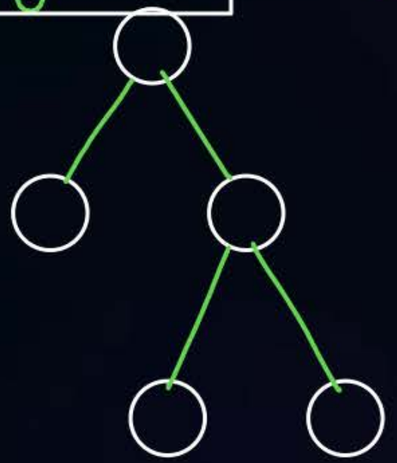


# Topic : Tree

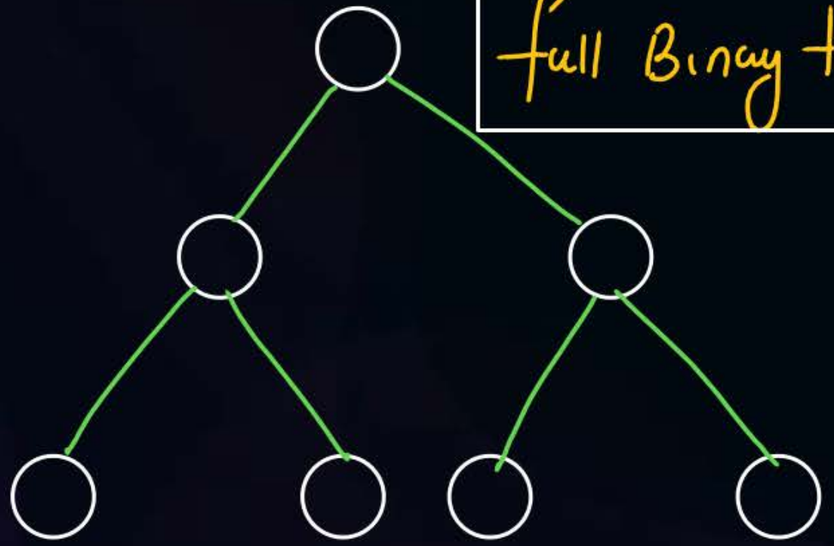
Binary tree :- A binay tree is a tree in which each node is having 0, 1 or 2 children.

2-ary tree

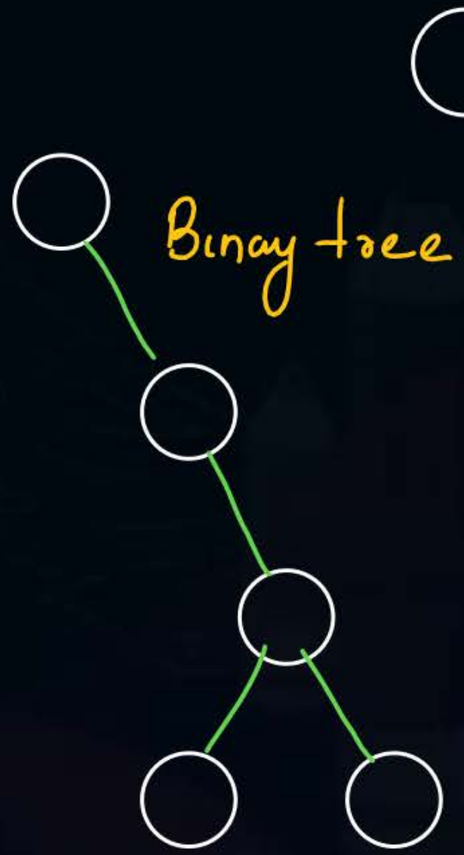
Each Node at most 2- children



full Binay tree



Binay tree



Skewed Binay tree



Slide

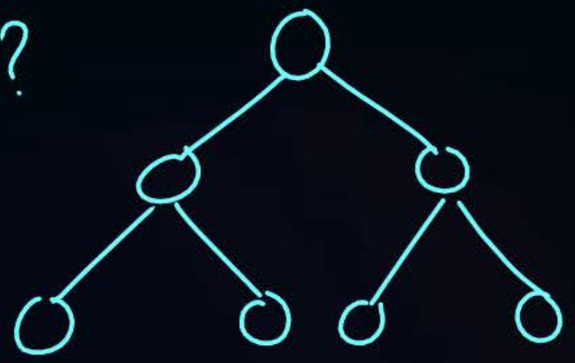


# Topic : Tree

full Binay tree : FBT is tree in evaylevel is full

2-ary tree?

yes



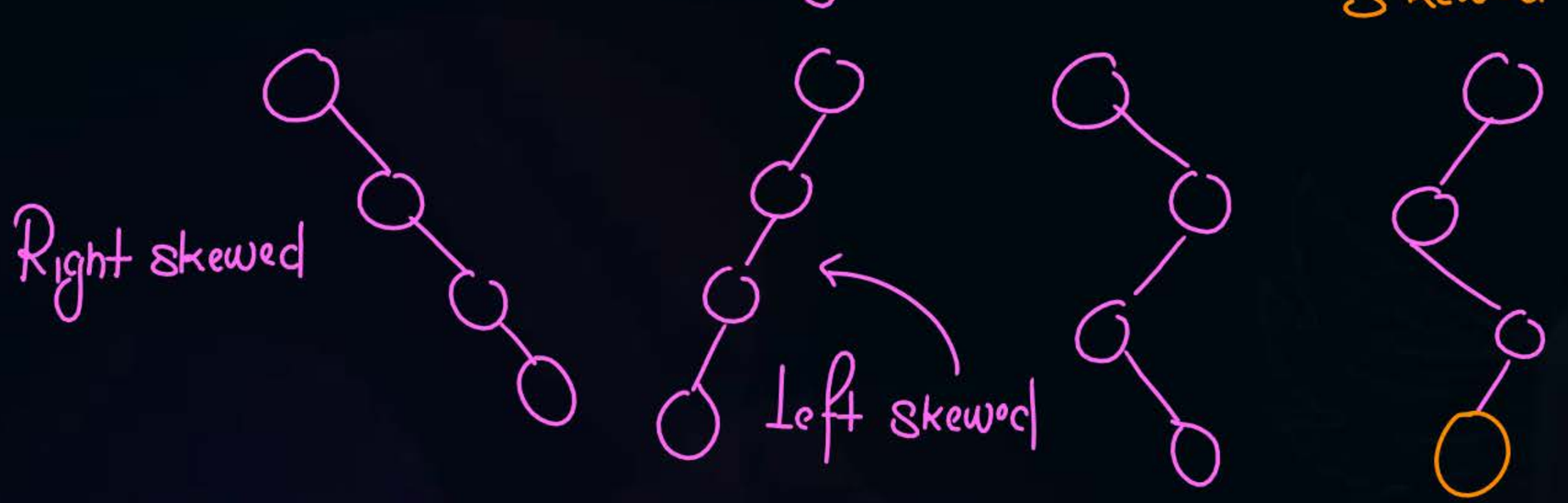
Minimum Height

Slide



# Topic : Tree

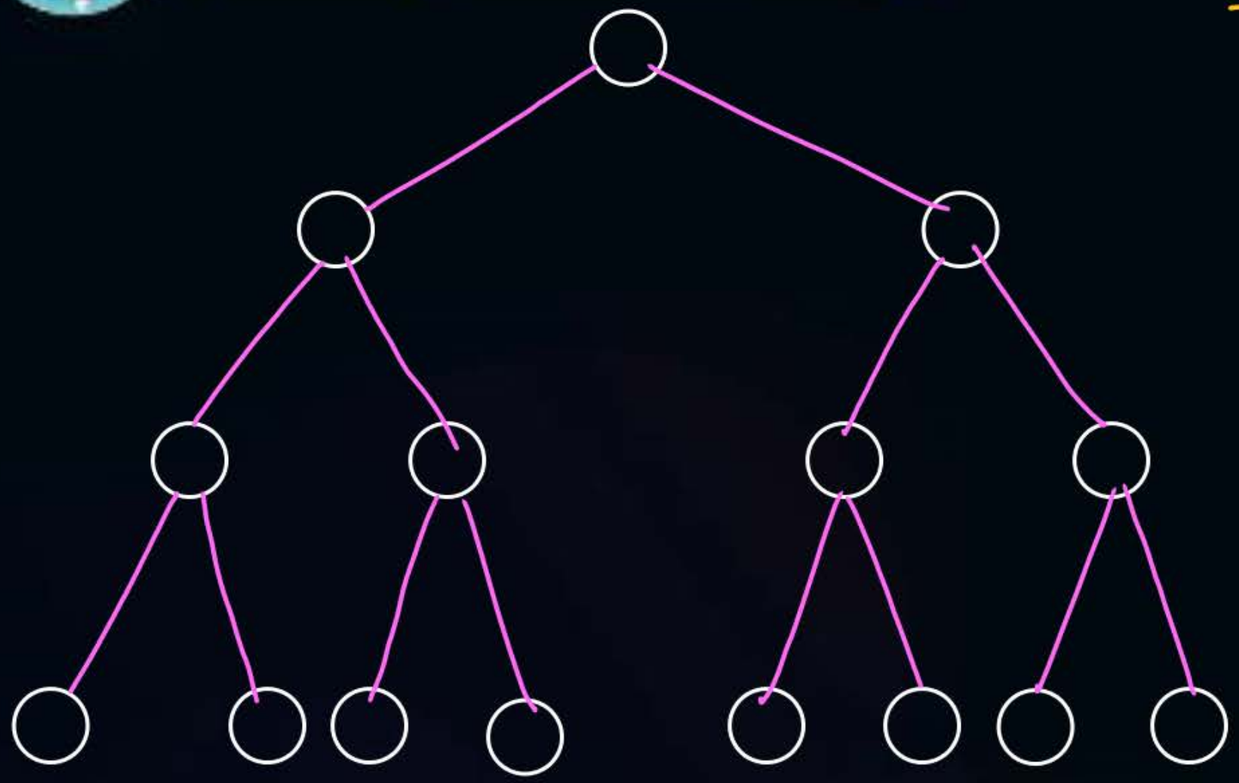
Skewed Binay tree :- Skewed binay tree is a tree in which each Node is having 1 children except 1 Node ( 1 Node has 0 - children)




Slide



# Topic : Tree



Suppose n is total No of Node & height of tree is h

height-0 ⟶ $2^0$

height -1 ⟶ $2^1$

height :2 ⟶ $2^2$

:

height - h ⟶ $2^h$

Slide



# Topic : Tree

$$2^0 + 2^1 + 2^2 + \cdots + 2^h = n$$

$$\frac{a(r^n - 1)}{r - 1}$$

$$\frac{1(2^{h+1} - 1)}{2 - 1} = n$$

Height 7 FBT

No of Nodes

$$n = 2^{7+1} - 1$$

$$= 2^8 - 1$$

$$= 256 - 1 = 255$$

$$\boxed{2^{h+1} - 1 = n}$$

$$2^{h+1} = n + 1$$

taking Log both side

$$\log_2 2^{h+1} = \log_2 (n+1)$$

$$h+1 \log_2 2 = \log_2 (n+1)$$

$$h + 1 = \log_2 (n+1)$$

$$\boxed{h = \log_2 (n+1) - 1}$$

$n = 15$

$$h = \log_2 (15+1) - 1$$

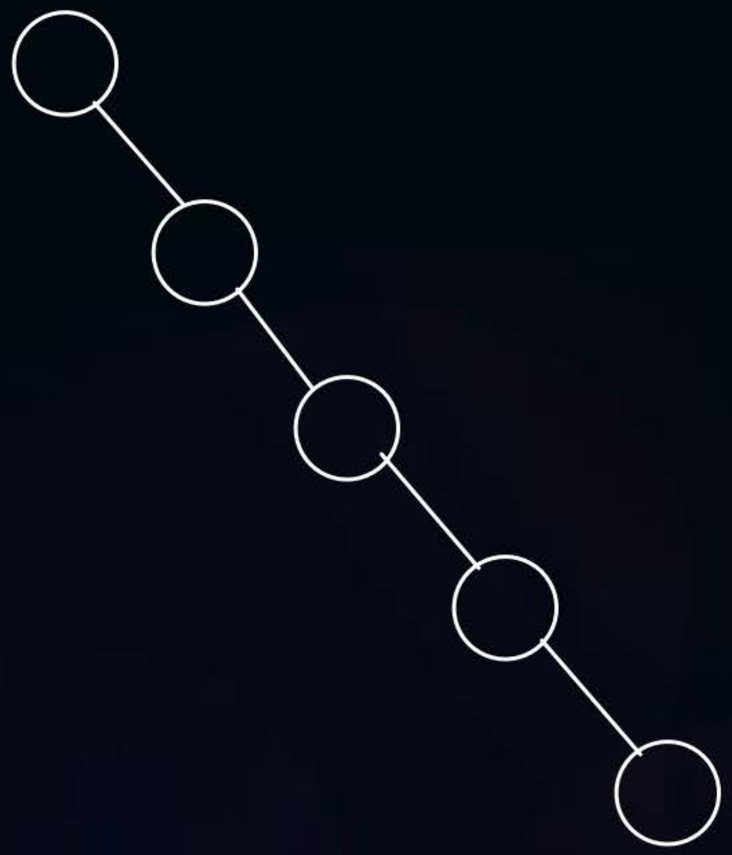
$$= \log_2 (16) - 1$$

$$= 4 - 1 = 3$$

Slide



# Topic : Tree



height - 0 ⟶ 1

height 1 ⟶ 1

height 2 ⟶ 1

height h ⟶ 1

$h + 1 = n$

$h = n - 1$

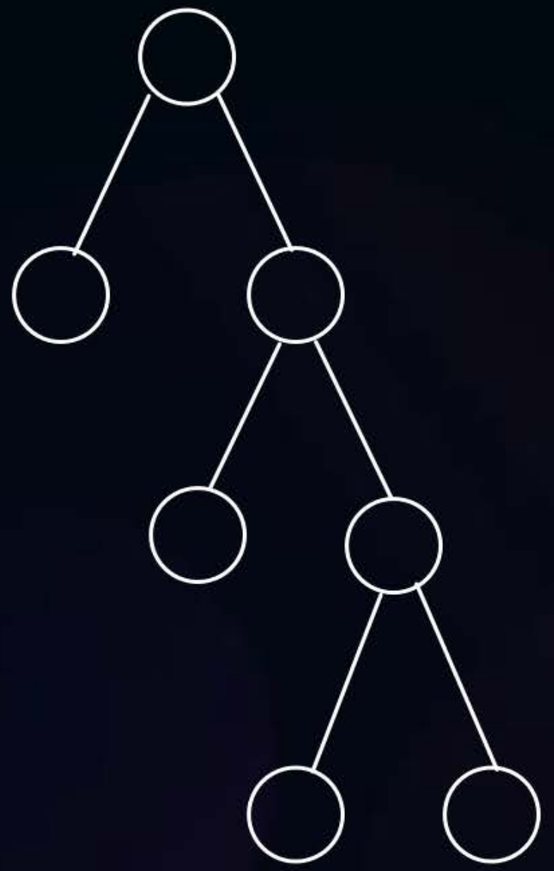
total No. of Nodes = $h + 1$

Slide



# Topic : Tree

#Q Minimum No of Nodes in k ary tree of Height h:



height - 0 — 1

height - 1 — k

height 2 — k

heigh - h — k

$h \times k + 1$

$n = h * k + 1$

$h = \frac{n-1}{k}$

Slide



# Topic : Tree

# Maximum No. of Nodes in the k-ary tree of height h

height 0 —— 1

height 1 —— k

height 2 —— $k^2$

⋮

height h —— $k^h$

total No of Nodes =

$1 + k + k^2 + \cdots k^h = n$

$$\frac{1(k^{h+1}-1)}{k-1} = n$$

$$n = \frac{k^{h+1}-1}{k-1}$$

Slide



# Topic : Question

#Q A 2-3 tree is tree such that

(A) all internal nodes have either 2 or 3 children ✓

(B) all paths from root to the leaves have the same length - FBT

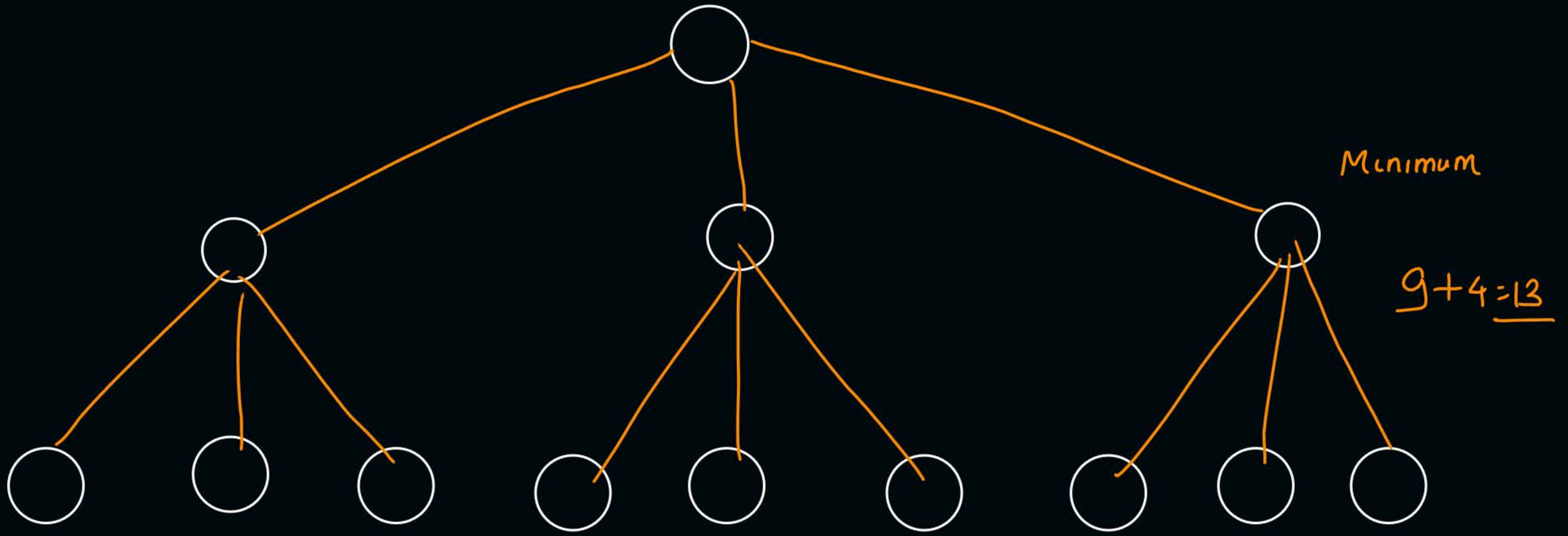
~~The minimum and maximum number of internal nodes of a 2-3 tree having 9 leaves could be~~

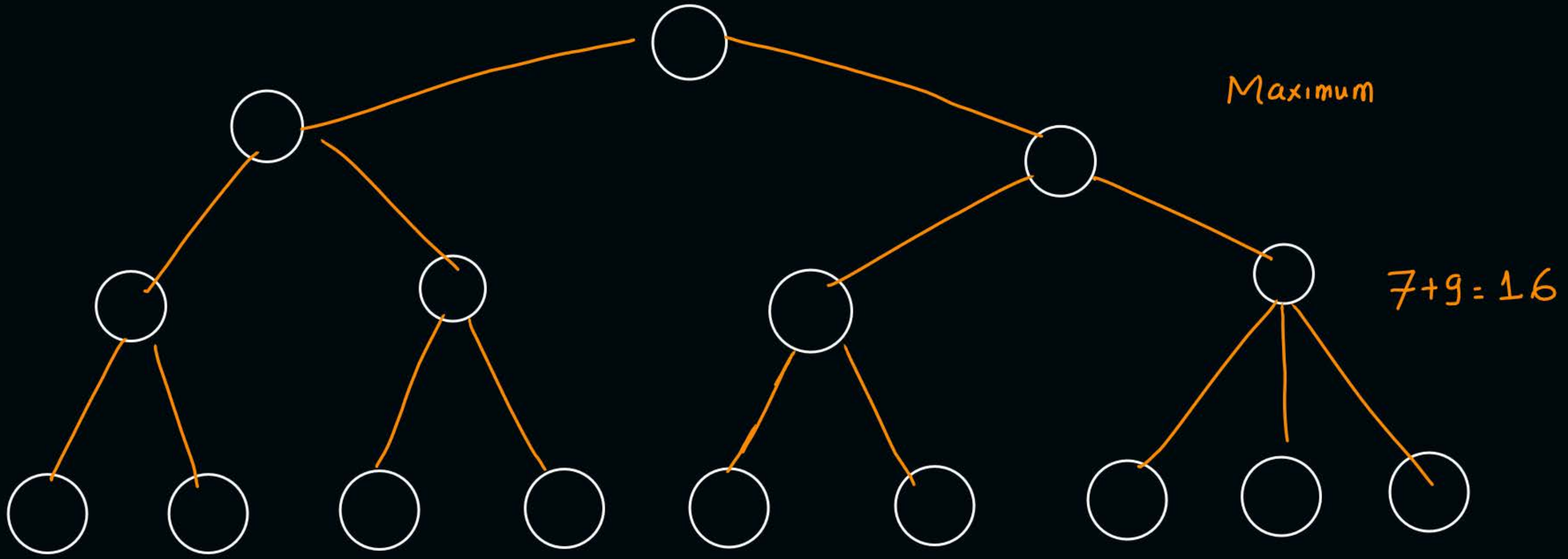
if mininmum No. of Nodes is x

maximum No. of Nodes is y

in 2-3 tree with 9 leaves then x+y is 29

Slide


Minimum
9+4=13


Maximum
7+9=16



# Topic : Question

#Q A complete n-ary tree is a tree in which each node has n children. Let l be the number of internal nodes and L be the number of leaves in a complete n-ary tree. If L = 41, and I = 10, what is the value of n?

(A) 3

(B) 4

(C) 5

(D) 6

Slide



# Topic : Question

#Q A complete k-ary tree is a tree in which each node has k children or no children. Let there be 24 internal node and 97 be the number of leaves. The value of the k is

$$L = I(k-1)+1$$

$$97 = 24(k-1)+1$$

$$24(I-1) = 96$$

$$I-1 = 4$$

$$I = 5$$

Slide



# Topic : Question

#Q Minimum number of nodes in 5-ary tree with height 6 is _____

k * h + 1

5 × 6 + 1 = 31

Slide



# Topic : Question

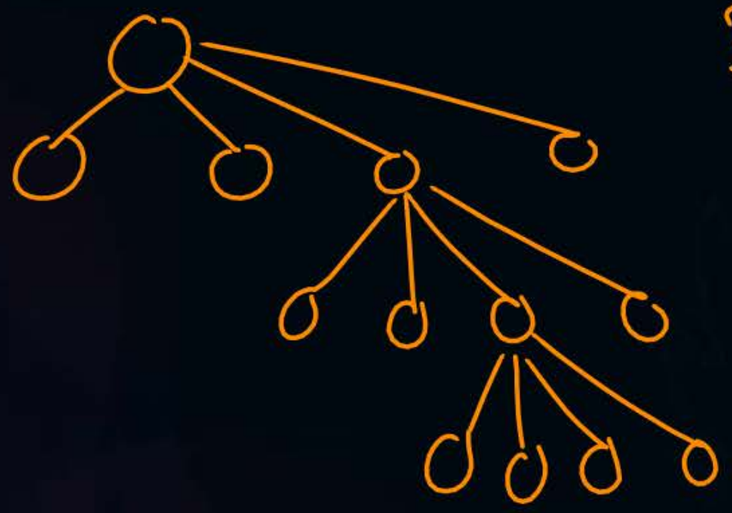
A full quaternary tree is a tree with every node having 0 or 4 children and all leafs are at same level. If there are n leaf nodes in a full quaternary tree at height h then how many internal nodes are present in the tree(root node at height 0)?

(A) $\frac{(n-1)}{2}$

(B) $\frac{(n-1)}{3}$

(C) $\frac{(n-1)}{4}$

(D) $\frac{(n-1)}{5}$



30% Maths + Aptitude

L = I(4-1)+

n =

$\frac{n-1}{3}$

Slide



# 2 mins Summary

Topic

k-ary tree

Topic

Minimum No. of Node / Maximum $k^{h+1}$

$$\frac{k^{h+1}-1}{k-1}$$

Topic

Topic

FBT heigh $h = \log_2(n+1) - 1$

Topic

Maximum Heigh $h = n-1$

Slide

# THANK - YOU

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